

Wavin AHC 9000

Modbus Communication Guide

Complete Reference for Modbus RTU Integration

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1. System Overview

What is the Wavin AHC 9000?

The Wavin AHC 9000 is an underfloor heating control system that communicates via Modbus RTU over RS-485. It controls up to **17 heating zones** (channels) with wireless thermostats and actuators.

Communication Settings

Parameter	Value
Protocol	Modbus RTU over RS-485
Baud Rate	38,400 bps
Data Format	8 data bits, 1 start, 1 stop, no parity
Default Address	0x01 (single unit)

2. What Can You Control?

Temperature Control for Each Room/Zone

- Set desired temperatures (Manual, Comfort, Eco, Standby, Party, Holiday)
- Read current air temperature
- Read current floor temperature
- Set temperature limits (min/max)
- Control heating/cooling mode

Domestic Hot Water (DHW)

- Enable/disable DHW function
- Set temperatures for different modes (Comfort, Eco, Standby, Cleaning)
- Schedule automatic cleaning cycles
- Monitor DHW sensor temperature

Scheduling

- Create weekly schedules for each zone
- 48 time slots per day (30-minute intervals)
- Different schedules for each day of the week
- Comfort/Eco switching based on schedule

Safety Features

- High temperature cut-off
- Freezing alarms
- Overcurrent protection
- Floor temperature limits

System Monitoring

- Monitor all connected thermostats
- Check battery status of wireless devices
- View signal strength (RSSI)
- Track power consumption
- Check device online/offline status

3. System Architecture

Data Organization Hierarchy

The system organizes data in a three-level structure: **CATEGORY** → **PAGE** → **REGISTER**

Example:

To read the temperature of Room 3:

- **Category:** PACKED DATA (0x02)
- **Page:** 3 (for Channel 3)
- **Register:** DESIRED TEMPERATURE (0x10)

Main Categories

Category	Name	Purpose	Pages
0x00	MAIN	Global system settings	1
0x01	ELEMENTS	Thermostat data	48
0x02	PACKED DATA	Room/zone settings	17
0x03	CHANNELS	Channel status	17
0x04	RELAYS	Relay control	2
0x05	CLOCK	Date/time	1
0x06	SCHEDULES	Weekly schedules	17
0x07	INFO	System information	1

4. Getting Started

Step 1: Connect to the System

- Wire RS-485 connection to the control unit
- Configure your Modbus master with the correct settings
- If multiple units exist, perform enumeration

Step 2: Read System Information

- Read the INFO category to identify your system
- Check control unit address
- Verify hardware and software version

Step 3: Discover Connected Devices

- Read the ELEMENTS category to find all connected thermostats
- Check which elements (thermostats) are present
- Verify battery and signal status

Step 4: Configure Channels

- Set temperature limits for each zone
- Configure operating mode
- Create schedules (optional)
- Link relays if needed

5. Main System Features

Global Settings (MAIN Category - 0x00)

DHW Temperature Settings

Setting	Register Index	Description
Comfort	0x14	Normal hot water temperature
Eco	0x15	Energy-saving temperature
Cleaning	0x16	Legionella prevention temperature
Standby	0x17	Minimal temperature when inactive

Temperature Format: Value $\times 0.1^{\circ}\text{C}$
Example: 450 = 45.0°C, 550 = 55.0°C

Monitor Temperatures

Sensor	Register Index	Description
DHW	0x0E	Current hot water temperature
Inlet	0x0F	Inlet water temperature

6. Room/Zone Control

Understanding Channels and Elements

- **Channel:** A heating zone (e.g., 'Living Room' = Channel 1)
- **Element:** A physical device (thermostat or sensor)
- **Room:** Multiple channels can be grouped together

You can have up to **17 channels** (Channels 1-16 for heating zones, Channel 17 special purpose).

Temperature Control (PACKED DATA Category - 0x02)

Each channel has its own page. Channel 1 uses Page 1, Channel 5 uses Page 5, etc.

Temperature Setpoints

Setting	Register	Description
Manual	0x00	User-set temperature
Comfort	0x01	Warm temp for occupied times
Eco	0x02	Lower temp for unoccupied times
Holiday	0x03	Long-term absence temperature
Standby	0x04	Minimal temperature
Party	0x05	Temperature boost for events

Note: All temperatures in 0.1°C units

Operating Modes

The mode is controlled by bits in the CONFIGURATION register (0x07):

Mode	SCHED_ENA	MODE[2:0]	Description
Manual	0	000	Fixed temperature
Permanent Standby	0	001	Always standby temp
Permanent Eco	0	010	Always eco temp
Permanent Comfort	0	011	Always comfort temp
Party (Manual)	0	100	Manual + party boost
Holiday (Manual)	0	101	Manual → holiday temp
Week Schedule	1	000	Follow programmed schedule
Temporary Eco	1	010	Override schedule → eco
Temporary Comfort	1	011	Override schedule → comfort
Party (Schedule)	1	100	Schedule + party boost
Holiday (Schedule)	1	101	Schedule → holiday temp

7. Scheduling

How Schedules Work

- **17 schedules** available (one per channel)
- **7 days** per week (Monday-Sunday)
- **48 time slots** per day (each = 30 minutes)
- Each slot can be **Comfort** (1) or **Eco** (0)

Schedule Structure (SCHEDULES Category - 0x06)

Each schedule occupies one page (0-16 for Channels 1-17).

Registers:

- 0x00: Header (schedule identifier)
- 0x01-0x15: Schedule data (21 registers = 3 per day × 7 days)

Creating a Schedule

Each day requires 48 bits (3 × 16-bit registers):

Time Slot	Bit Number
00:00-00:30	Bit 0
00:30-01:00	Bit 1
01:00-01:30	Bit 2
...	...
23:30-24:00	Bit 47

8. Practical Examples

Example 1: Set Living Room to 21°C (Manual Mode)

Step 1: Write temperature setpoint

```
Category: 0x02 (PACKED DATA)
Page: 1 (assuming Living Room = Channel 1)
Register: 0x00 (MANUAL TEMPERATURE)
Value: 210 (= 21.0°C)
```

Step 2: Set mode to Manual

```
Category: 0x02 (PACKED DATA)
Page: 1
Register: 0x07 (CONFIGURATION)
Set SCHED_ENA = 0, MODE = 000
```

Example 2: Enable DHW with 55°C Comfort Temperature

Step 1: Enable DHW

```
Category: 0x00 (MAIN)
Page: 0
Register: 0x08 (STATUS L)
Set bit 13 = 1
```

Step 2: Set comfort temperature

```
Category: 0x00 (MAIN)
Page: 0
Register: 0x14 (DHW COMFORT TEMPERATURE)
Value: 550 (= 55.0°C)
```

Example 3: Read Current Temperature in Bedroom

If you only know the channel (e.g., Channel 3):

```
Step 1: Find primary element
Category: 0x03 (CHANNELS)
Page: 3
Register: 0x02 (PRIMARY ELEMENT)
Read: Element index (e.g., 5 means Element Page 4)
```

```
Step 2: Read temperature
Category: 0x01 (ELEMENTS)
Page: 4
Register: 0x04 (AIR TEMPERATURE)
```

9. Quick Reference Tables

Common Register Addresses - Global Settings

Function	Cat	Page	Reg	Notes
Enable DHW	0x00	0	0x08	Bit 13
DHW Temp (Comfort)	0x00	0	0x14	× 0.1°C
DHW Temp (Eco)	0x00	0	0x15	× 0.1°C
Current DHW Temp	0x00	0	0x0E	Read only
Inlet Temperature	0x00	0	0x0F	Read only
Total Power	0x00	0	0x10-0x11	32-bit
System Time (Hour)	0x05	0	0x04	0-23
System Time (Min)	0x05	0	0x05	0-59

Per-Channel Settings (Page = Channel Number)

Function	Category	Register	Notes
Manual Temp	0x02	0x00	× 0.1°C
Comfort Temp	0x02	0x01	× 0.1°C
Eco Temp	0x02	0x02	× 0.1°C
Mode/Config	0x02	0x07	See mode table
Current Status	0x03	0x00	Output on/off, alarms
Power Usage	0x03	0x01	× 0.54 mA
Schedule	0x06	0x00-0x15	22 registers

Per-Thermostat Data (Page = Element Index)

Function	Category	Register	Notes
Air Temperature	0x01	0x04	Read only, × 0.1°C
Floor Temperature	0x01	0x05	Read only, × 0.1°C
Battery %	0x01	0x0A	0-10 (× 10%)
Online Status	0x01	0x08	Bit 15
Signal Strength	0x01	0x09	RSSI

Temperature Conversion

Reading: Divide register value by 10

- Register = 215 → 21.5°C
- Register = 450 → 45.0°C

Writing: Multiply desired °C by 10

- Want 20.5°C → Write 205
- Want 55.0°C → Write 550

Modbus Function Codes

Code	Name	Purpose
0x43	Read Register from Index	Read by Category/Page/Index
0x44	Write Register to Index	Write by Category/Page/Index
0x45	Write Register Masked	Write specific bits only
0x41	Read Register from Address	Read element by address
0x42	Write Register from Address	Write element by address
0x6D	Enumeration Commands	Setup multiple units

Important Notes

Temperature Values

- Always stored as integer $\times 10$
- Example: $22.5^{\circ}\text{C} = 225$
- Unknown temperature = $0x7FFF$

Bit Numbering

- Bit 0 = Least significant bit (rightmost)
- Bit 15 = Most significant bit (leftmost)

Writing Schedules

- Must write entire page at once (all 22 registers)
- Partial writes may cause malfunction

Multiple Units

- Default address: $0x01$ (works for single unit)
- Multiple units require enumeration process
- Each unit gets unique logical address ($0x02$ - $0xF7$)

Safety

- Always configure temperature limits
- Enable high-temperature cut-off for safety
- Monitor frost alarms in cold climates

Summary

The Wavin AHC 9000 Modbus interface provides comprehensive control over:

- ✓ 17 heating zones with individual temperature control
- ✓ Domestic hot water management
- ✓ Weekly scheduling with 30-minute resolution
- ✓ Safety features (high-temp cut-off, freeze protection)
- ✓ Power monitoring for each zone
- ✓ Wireless device monitoring (battery, signal, status)
- ✓ Relay control for pumps and boilers

The system uses a hierarchical structure (Category → Page → Register) that makes it easy to organize and access thousands of data points across multiple zones and devices.

This guide is based on the Wavin Modbus Register Map for AHC 9000 control unit (AC-116)